

Following completion of the preparation materials, students should be able to:

1. PK properties:

Compare the class pharmacokinetic properties of the indirect-acting cholinomimetics to those of the individual agents, including the tendency to cross the blood-brain barrier.

2. Structure-activity relationship → Mechanism of action and effects:

Correlate the chemical structures of cholinesterase inhibitors with their mechanisms of action, duration of action, and cholinomimetic effects (the quaternary alcohol, carbamate esters, and organophosphates cholinesterase inhibitors).

3. Cholinomimetic effects → Therapeutic uses

Relate the effects of the cholinesterase inhibitors to their therapeutic effects and diagnostic uses.

4. Cholinomimetic effects → Toxicities and cautions for use:

Predict the signs and symptoms of cholinesterase toxicity based on the mechanisms of action and effects of the drugs.

5. Poisoning antidotes:

Explain the actions and effects of the antidotes for cholinesterase toxicity.

Key points: Physiology-Pharmacology Relationship

- ***Understanding acetylcholinesterase's mechanism of action is the key to understanding the actions of the cholinesterase inhibitors.***
- ***Understanding acetylcholine's effects on the nicotinic and muscarinic receptors in the CNS, the periphery, and the neuromuscular junction is the key to understanding the effects of the cholinesterase inhibitors.***
- Cholinesterase inhibitors block the actions of acetylcholinesterase, which increases acetylcholine concentrations in the synaptic clefts and neuroeffector junctions.
- The excess acetylcholine, in turn, stimulates cholinceptors and evokes increased responses – **cholinesterase inhibitors are amplifiers of endogenous acetylcholine.**
- Cholinesterase inhibitors also inhibit butyrylcholinesterase (BChE), which is a nonspecific enzyme that hydrolyzes many different choline esters. Other names are pseudocholinesterase and plasma cholinesterase. Note: BChE is not involved in the physiologic termination of acetylcholine.

Key points: Structure-Activity Relationship

Cholinesterase inhibitors are subcategorized according to their chemical structures, which determine their durations of action.

- Simple alcohol: Edrophonium has a quaternary ammonium cation and short duration of action.
- Carbamic acid esters: Neostigmine, pyridostigmine, and physostigmine are reversible cholinesterase inhibitors with durations of action of several hours.
 - Neostigmine and pyridostigmine are synthetic compounds consisting of a quaternary ammonium cation (like acetylcholine) and a carbamate group (instead of a $-\text{COOH}$ group). Predictably, they are poorly absorbed and distribution to the CNS is negligible.
 - Physostigmine is a naturally occurring alkaloid, a tertiary amine, that is well absorbed and distributes to the CNS due to its greater lipid solubility.
- Organophosphates (organic derivatives of phosphoric acid) are irreversible cholinesterase inhibitors with very high lipid solubility. They were developed as insecticides that would be absorbed and distributed to the insect's CNS very rapidly. Several nerve toxins, such as sarin gas, are extremely toxic to humans.

Key points: Mechanisms of action: Inhibiting AChE increases ACh levels which enhances ACh actions at nicotinic and muscarinic receptors.

To understand the mechanism of action of the cholinesterase inhibitors, it is necessary to understand the mechanism of acetylcholine hydrolysis:

- Acetylcholine (ACh) binds to acetylcholinesterase (AChE) at two sites – the ammonium cation at the anionic site and the carboxyl group at catalytic triad of the esteratic site.
- Hydrolysis of acetylcholine releases acetate, choline, and the active enzyme.

The interactions of the three subgroups of cholinesterase inhibitors differ according to their structures:

- The simple alcohol, edrophonium, reversibly binds to the anionic site, preventing access of ACh. It's duration of action is minutes.
- The carbamate esters undergo the same process as ACh. However, the carbamoylated enzyme is more resistant to hydrolysis. The duration of action is several hours.
- The organophosphates irreversibly phosphorylate the enzyme forming an extremely stable bond. The phosphorylated enzyme undergoes aging that further strengthens the bond.

Key points: Therapeutic Uses

System / Drugs	Therapeutic uses	Mechanism
Skeletal muscle Neostigmine, Pyridostigmine	Nondepolarizing neuromuscular blocker reversal agent Myasthenia gravis treatment	Enhance ACh levels in the neuromuscular junction Direct cholinomimetic effect on skeletal muscle (and indirect and direct effects on autonomic ganglia)
Urinary tract Neostigmine	Acute colonic pseudo-obstruction (Ogilvie syndrome)	Promotes bladder emptying by contraction of the trigone (smooth muscle)
Eye Echothiophate	Accommodative esotropia (strabismus) Reduce intraocular pressure	Ciliary muscle stimulation, potentiate accommodation, facilitate aq. humor outflow
CNS Donepezil Rivastigmine Galantamine	Alzheimer disease symptomatic treatment	Appear to increase central ACh in the basal forebrain and cortex that contribute to cognitive deficits
CNS Physostigmine	Reversal of CNS anticholinergic syndrome	Increased CNS ACh synaptic levels outcompetes anticholinergic agents at cholinergic receptors

Key Points: Adverse effects predictably result from cholinomimetic actions (stimulation) on muscarinic and nicotinic receptors.

Cardiovascular, gastrointestinal, eye, and skeletal muscle cholinergic effects are most prominent effects of cholinesterase inhibitors.

Effect	System
muscle cramps / twitching muscle weakness	Skeletal muscle
miosis, visual disturbance	Eye, facilitated accommodation
bradycardia, AV nodal block, arrhythmia	Cardiac
hypotension with reflex tachycardia syncope flushing	Vascular
bronchospasm	Lung
salivation, lacrimation airway secretions GI secretions profuse sweating	Glandular secretions
abdominal cramps, borborygmi, nausea, vomiting, diarrhea	GI smooth muscle and secretions
urinary urgency / incontinence	Stimulation of micturition

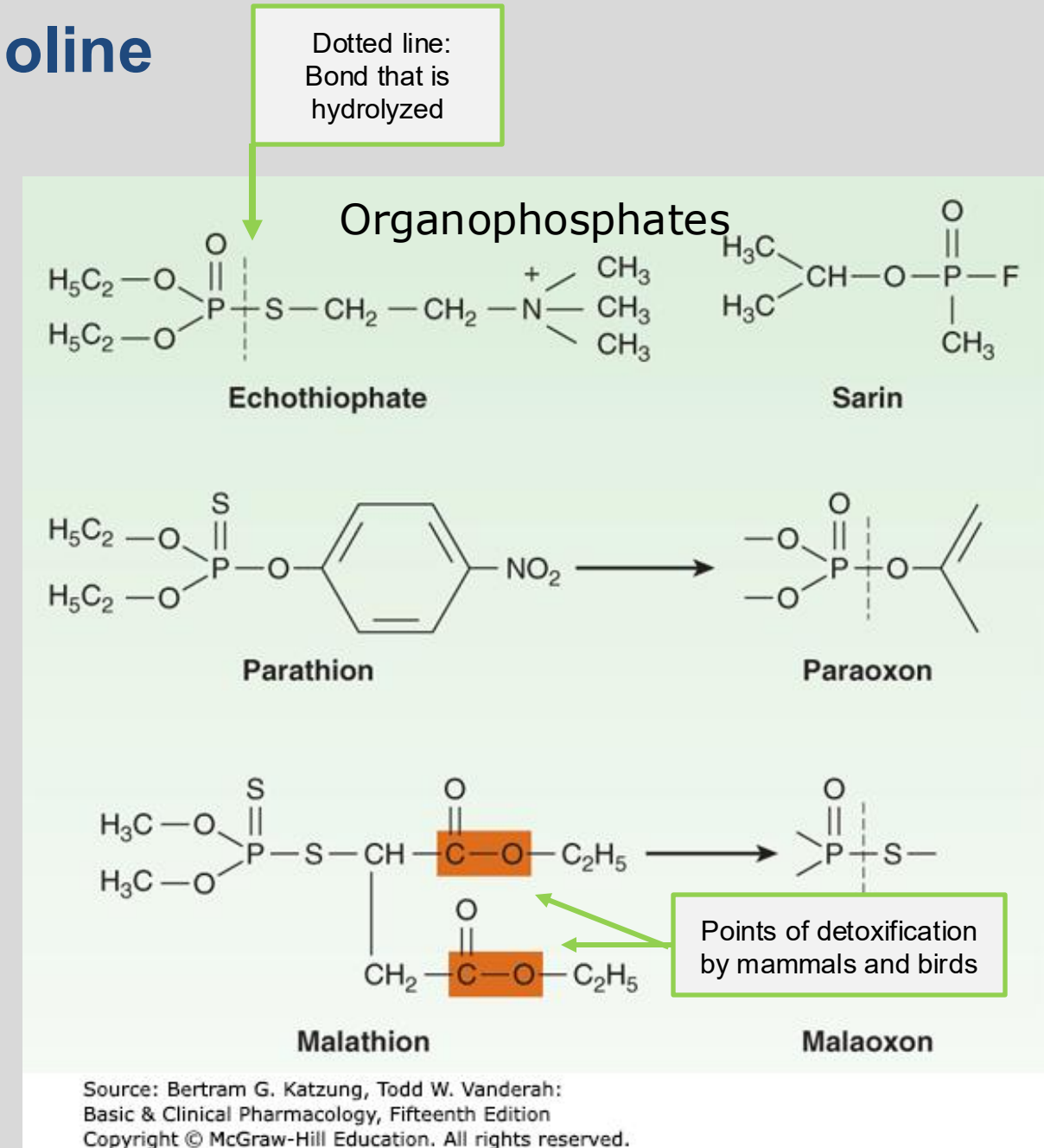
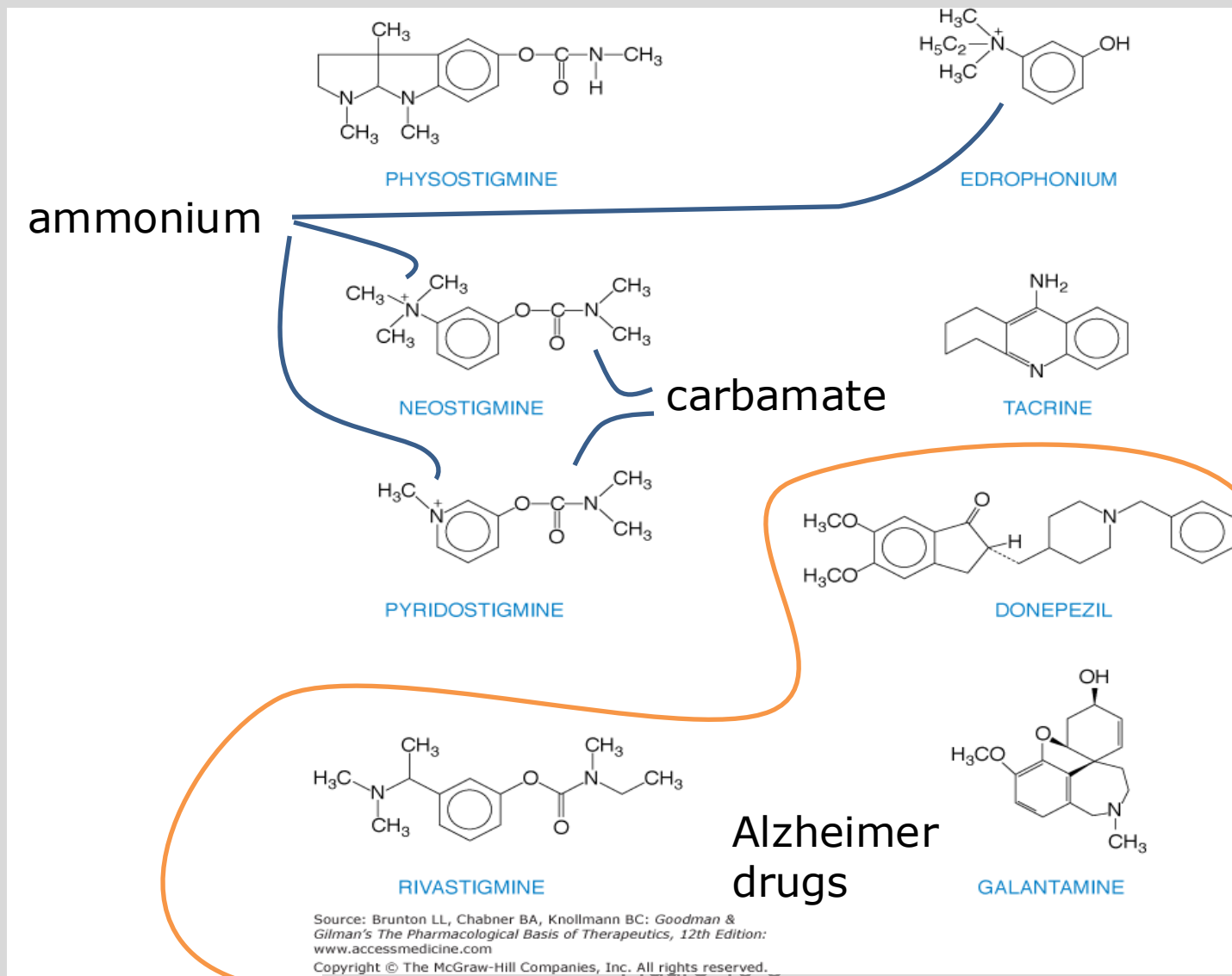
Cautions

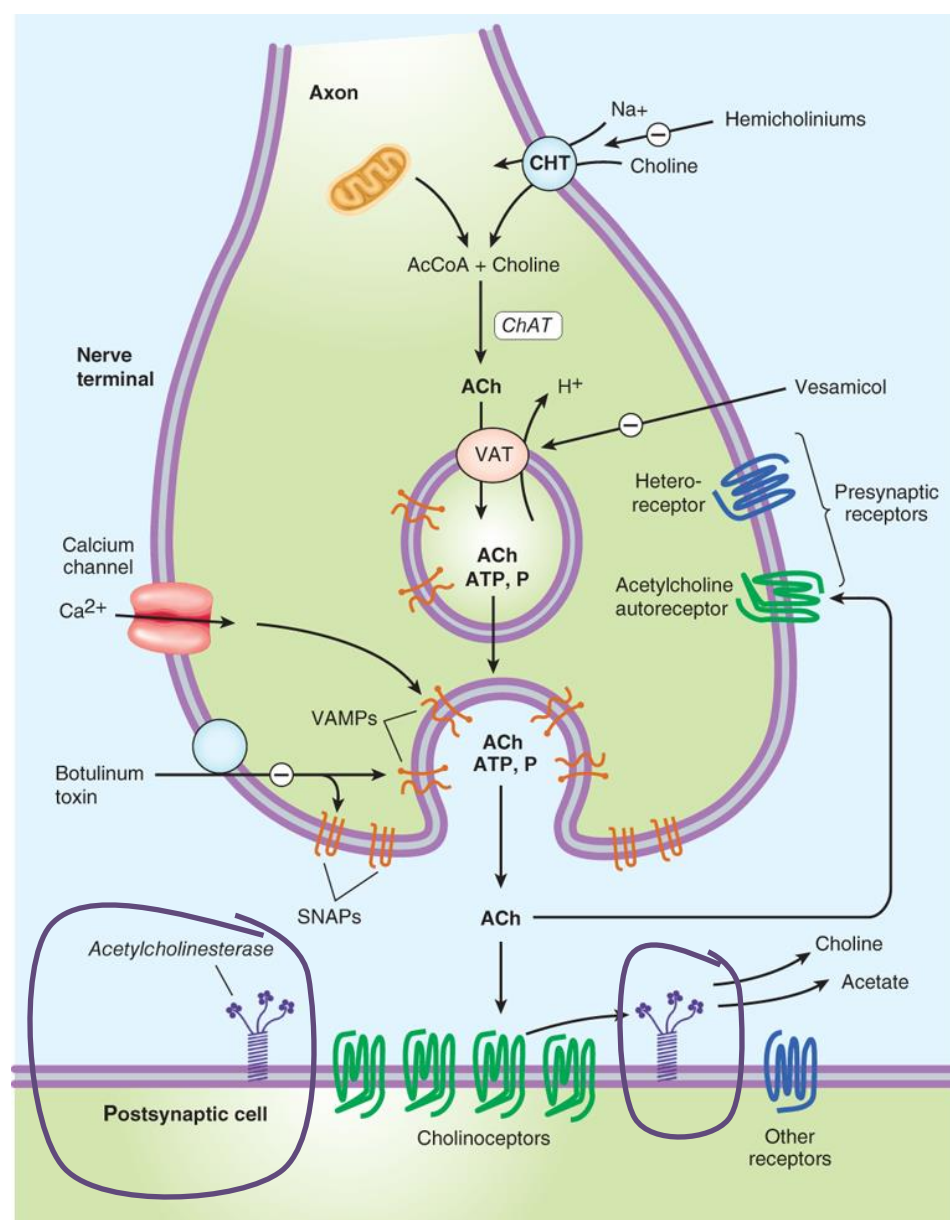
- muscle weakness with excessive dose
- miosis may reduce night vision
- cardiovascular disease
 - bradycardia (risk arrhythmia)
 - heart block (slow conduction, AV node)
 - hypotension (→ reflex tachycardia)
 - ischemic heart disease (risk angina)
 - hyperthyroidism (risk atrial fibrillation)
- asthma
- chronic obstructive pulmonary disease
- peptic ulcer disease
- gastrointestinal or urinary bladder surgery, resection or obstruction
- Drugs that enhance adverse effects or interfere with the actions of anticholinergic drugs

Cholinesterase inhibitors

• Indirect-acting cholinomimetics

Agents that amplify the actions of acetylcholine





Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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1. Choline transport
2. $\text{Ch} + \text{AcCoA} = \text{ACh}$
3. VAT → Transport of ACh into storage vesicle
4. Axon terminal depolarization → Ca^{2+} influx
5. Vesicle attachment and fusion
6. Exocytosis of ACh & cotransmitters into junctional cleft
7. $\text{AChE} \rightarrow \text{Ch} + \text{acetate} \rightarrow \text{terminate action}$
8. Choline recycled; acetate diffuses away

Schematic illustration of a generalized cholinergic junction (not to scale). Choline is transported into the presynaptic nerve terminal by a sodium-dependent choline transporter (CHT). This transporter can be inhibited by hemicholinium drugs. In the cytoplasm, acetylcholine is synthesized from choline and acetyl-CoA (AcCoA) by the enzyme choline acetyltransferase (ChAT). Acetylcholine (ACh) is then transported into the storage vesicle by a vesicle-associated transporter (VAT), which can be inhibited by vesamicol. Peptides (P), adenosine triphosphate (ATP), and proteoglycan are also stored in the vesicle. Release of transmitters occurs when voltage-sensitive calcium channels in the terminal membrane are opened, allowing an influx of calcium. The resulting increase in intracellular calcium causes fusion of vesicles with the surface membrane and exocytotic expulsion of acetylcholine and cotransmitters into the junctional cleft (see text). This step can be blocked by botulinum toxin. Acetylcholine's action is terminated by metabolism by the enzyme acetylcholinesterase. Receptors on the presynaptic nerve ending modulate transmitter release. SNAPs, synaptosomal nerve-associated proteins; VAMPs, vesicle-associated membrane proteins.

Acetylcholinesterase Inhibitors: Actions and Effects

Indirect-acting cholinomimetics

inhibit the hydrolytic action of acetylcholinesterase



Acetylcholine accumulates

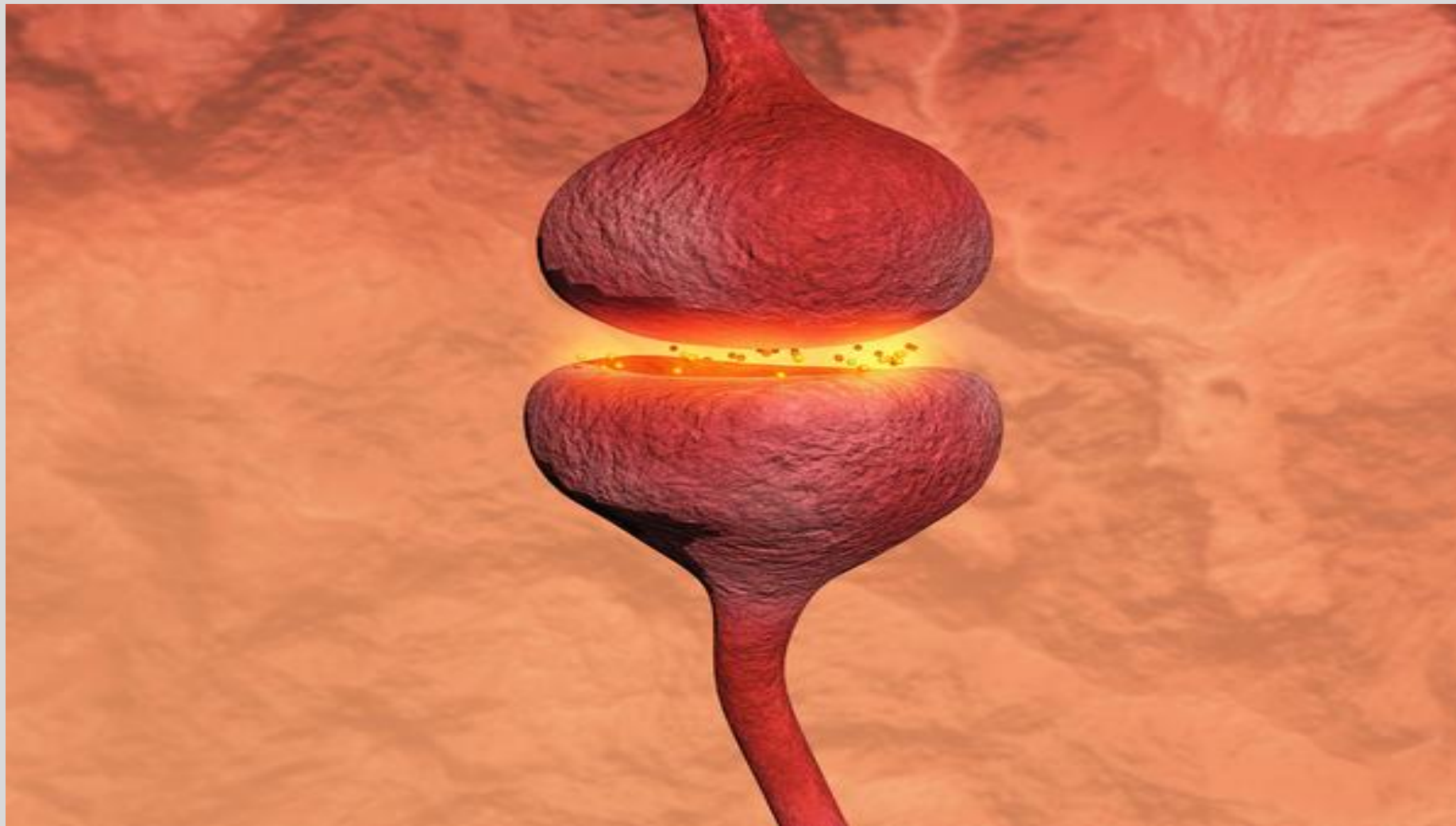
in the vicinity of cholinergic nerve terminals



Enhances/Augments acetylcholine activity

Clinical pearl: Accumulation increases the potential for stimulation – even excessive stimulation – of cholinergic receptors throughout the central and peripheral nervous systems.

Preventing the breakdown of acetylcholine
increases the level and duration of acetylcholine action.



- Why are cholinesterase inhibitors called indirect-acting cholinomimetics?
- What are the effects of inhibiting the actions of cholinesterase?

Mechanism: Acetylcholinesterase inactivation of acetylcholine

Step 1

Ammonium binds to anionic site.
Carboxyl group binds serine
in the catalytic triad
of the esteric site.

Step 2

Catalytic Step

enzyme cleaves choline
from ACh

Free choline

enzyme-acetate intermediate

Step 3

Hydrolysis Step

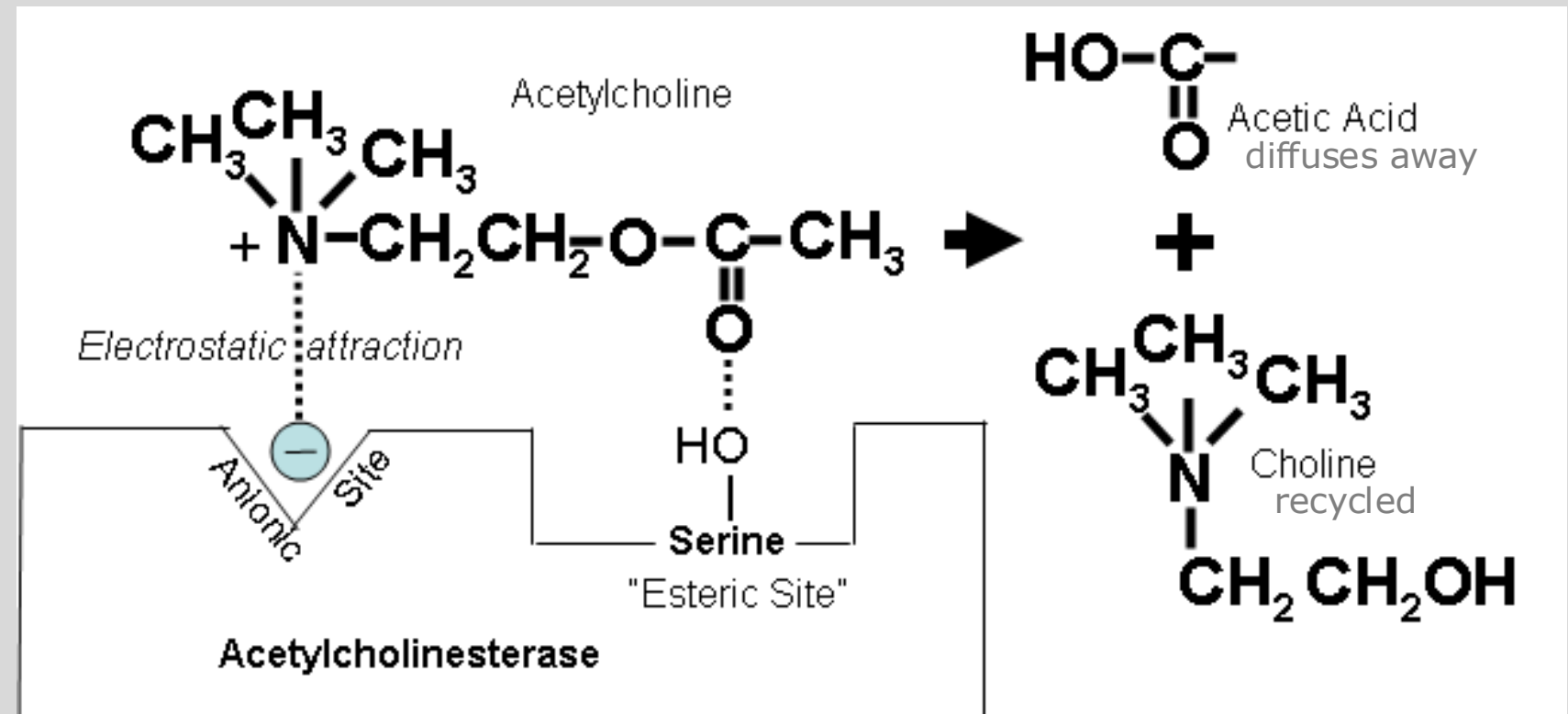
acetate couples to the
esteratic site of the enzyme

H₂O
Bond is split

acetate + active enzyme

Hydrolysis of ACh releases:

choline + acetic acid + active enzyme & termination of ACh action

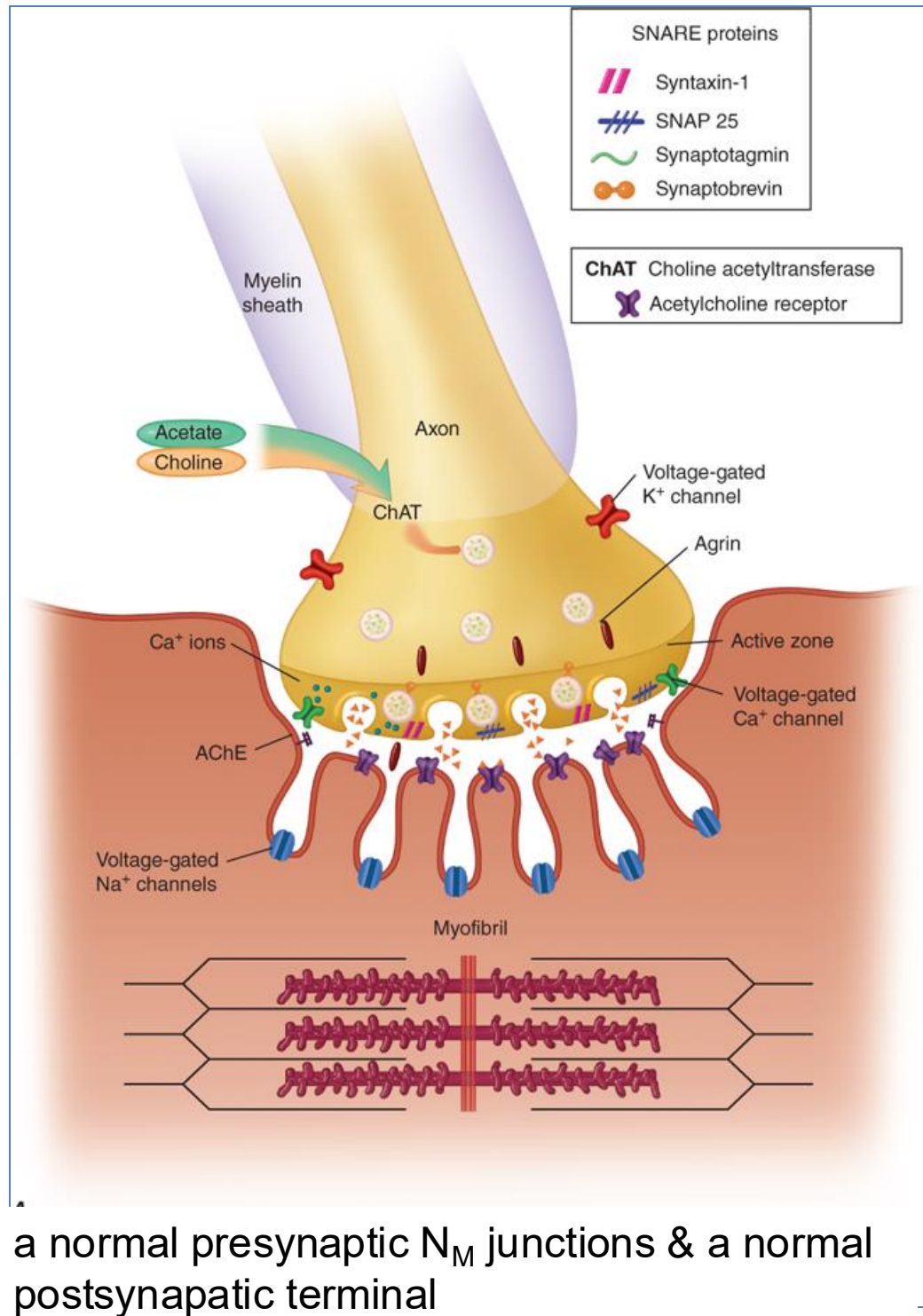


Inactivation of ACh by AChE takes about 150 microseconds.

Cholinesterase Inhibitors Mechanism of Action

Quaternary alcohol	Competitive Inhibition - reversibly binds electrostatically and by hydrogen bonds to the active site - rapid dissociation → short duration of action (2-10 min)	
Carbamic acid esters	Carbamoylation - Ammonium binds anionic site, carbamate binds esteratic site - slow hydrolysis → ↑ duration of action (30 min – 6 h)	
Organophosphates	Phosphorylation - covalent phosphorylation of the enzyme - extremely stable bond; hydrolyzes very slowly → essentially irreversible - Aging: further strengthening of phosphorus-enzyme bond → irreversible	
Cholinesterase inhibitors for symptomatic treatment of Alzheimer disease (pharmacology from slide 29)	Reversible cholinesterase inhibitors - Donepezil: noncarbamate - Galantamine: noncarbamate - Rivastigmine: carbamate	Cross blood-brain barrier - noncompetitive ChE inhibitor - competitive ChE inhibitor - carbamoylation of ChE

Drug	Uses	Duration of Action
Edrophonium no longer available	Dx of Myasthenia Gravis (historically) Edrophonium had suboptimal sensitivity and specificity. The most common test is serology to detect AChR autoantibodies.	~ 10 min (IV)
Neostigmine	Reversal of nondepolarizing neuromuscular blockade Myasthenia gravis treatment Acute colonic pseudo-obstruction (Ogilvie syndrome)	2.5-4 hrs
Pyridostigmine	Myasthenia gravis treatment Reversal of nondepolarizing neuromuscular block	6-8 hrs
Neostigmine and pyridostigmine (1) <i>inhibit cholinesterase</i> and (2) have <i>direct nicotinic agonist effect</i> at the N _M junction and autonomic ganglia.		
Physostigmine	Antidote for anticholinergic toxicity Reversal of toxic, life-threatening delirium caused by anticholinergic agents (e.g. atropine, jimson weed). Toxicities → cautious use: Physostigmine can cause bradycardia, respiratory distress, and seizures with too rapid administration.	45-60 min
Malathion	Topical lotion: head and pubic lice and ova, scabies	Lotion must remain on skin 8-12 h
Echothiophate	Ophthalmic drops: miotic, reduction of intraocular pressure	Long duration



Myasthenia Gravis pathophysiology

Autoimmune response



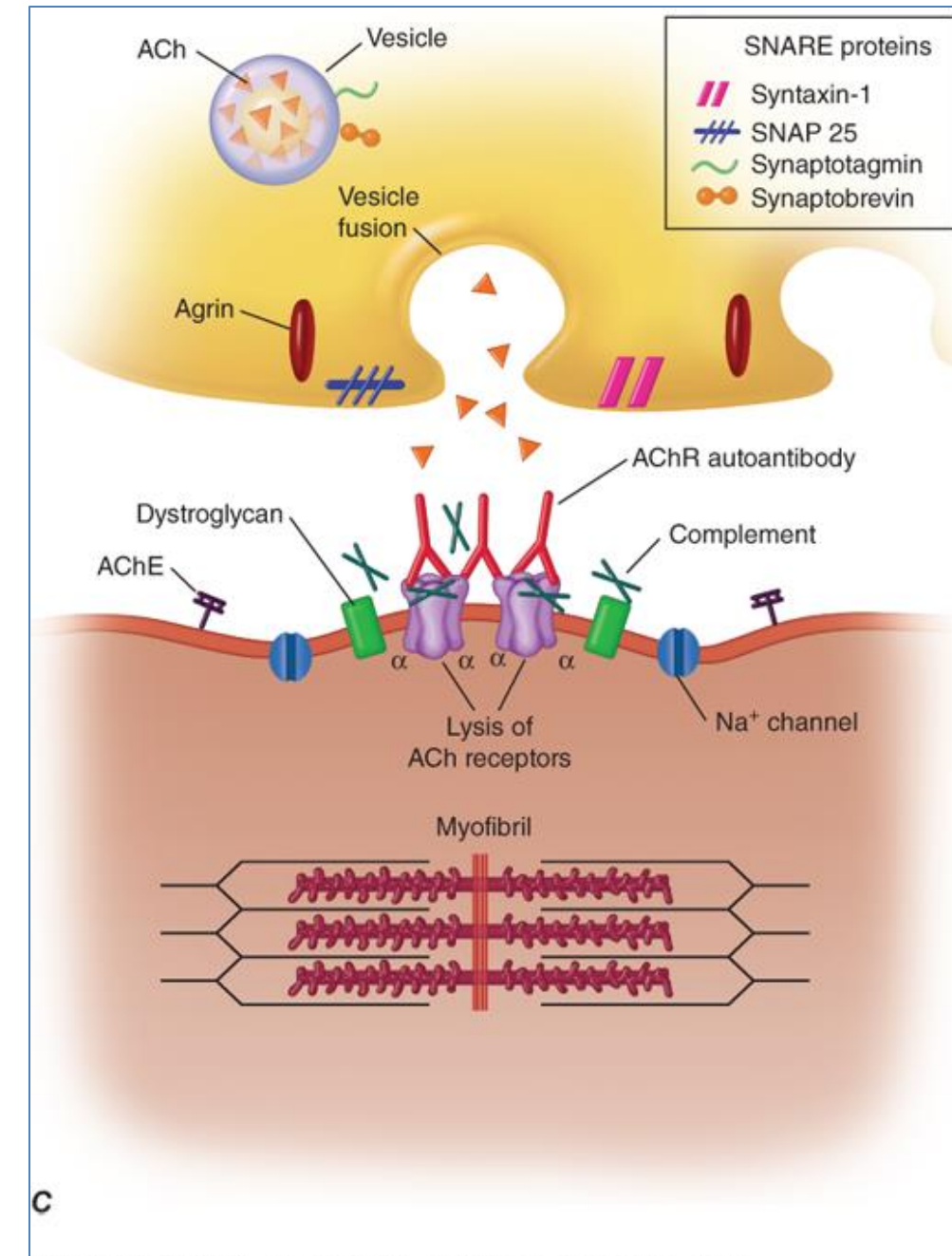
anti-nAChR antibodies



cross-linking & lysis of nAChRs which decreases the number of receptors at the N_M junction



failure to trigger muscle action potential (end-plate potential)



normal nerve terminal; ↓ number of nAChRs (stippling); flattened, simplified postsynaptic folds; and a widened synaptic space

Mention:

Edrophonium test for differentiating myasthenic crisis and cholinergic crisis is no longer performed but **it may still be on the board exams**.

Definitions:

- Myasthenic crisis: Life-threatening respiratory failure due to weakness of respiratory muscles.
- Cholinergic crisis: Excessive cholinesterase inhibitor intake overstimulates nicotinic receptors causing depolarizing block resulting in muscle weakness and respiratory failure.

Cholinergic crisis occurs very rarely if at all in patients taking pyridostigmine at therapeutic doses.

- **Practice point:** Detailed history and physical examination are necessary to diagnose cholinergic crisis. A history of exposure to cholinesterase inhibitors is key to diagnosing cholinesterase inhibitor toxicity.

The Test: A small dose of edrophonium → brief block of AChE in the N_M junction
(1) Improvement of muscle weakness is a positive indicator of myasthenia gravis.
(2) Muscle weakness does not resolve or worsens in cholinergic crisis.

Check your knowledge

- A patient who had surgery under general anesthesia facilitated by a nondepolarizing neuromuscular blocker is administered a drug to reverse the neuromuscular blockade. What is the mechanistic rationale for this approach?
- A patient diagnosed with myasthenia gravis is given treatment with an oral drug. What is the drug and what is the clinical rationale for this approach?

1) Atropine: Antimuscarinic Agent

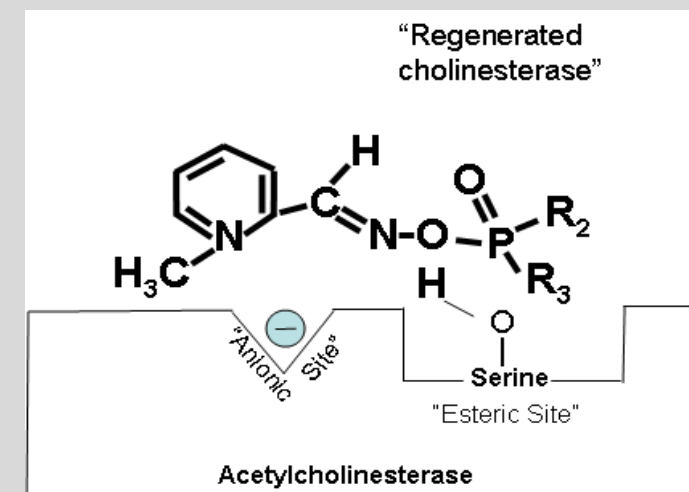
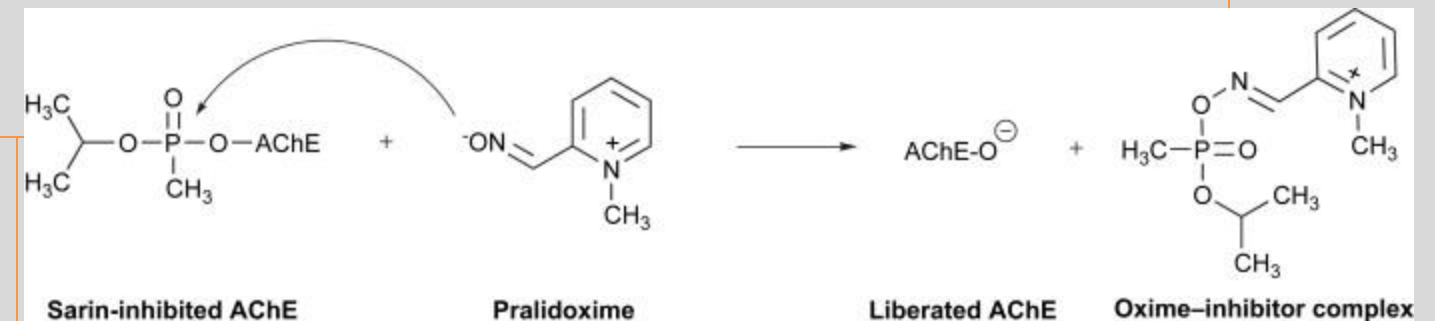
2) Pralidoxime (2-PAM): Cholinesterase Regenerator

Atropine

- Competitive antagonist at muscarinic receptors
- ***Reduces the symptoms of parasympathetic overstimulation*** resulting from excess acetylcholine in the synapses
- **Peripheral and CNS effects**

Pralidoxime

- Strong nucleophile
- High affinity for phosphorus atom in organophosphate
- Breaks the phosphorus-enzyme bond
- **Peripheral effects only**



Pralidoxime breaks the bond IF administered before aging occurs.

Interaction: <https://www.sciencedirect.com/topics/neuroscience/pralidoxime>

Diagram: Agency for Toxic Substances and Disease Registry Environmental Health and Medicine Education; Cholinesterase inhibitors
<http://www.atsdr.cdc.gov/csem/csem.asp?csem=11&po=0>

Check your knowledge

- The carbamate, organophosphate, Alzheimer's cholinesterase inhibitors have myriad, predictable, mechanism-related adverse effects. What systems are involved?

Centrally-Acting Cholinesterase Inhibitors for the symptomatic treatment of Alzheimer disease

- CNS actions and effects for the symptomatic treatment of Alzheimer disease:
 - Cholinergic neurons project to various cortical areas. These neurons control attention, thinking, and processing of stimuli. They span the forebrain and encompass the brainstem and thalamus, which are responsible for consciousness and attention.
 - In Alzheimer dementia, these cholinergic neurons have defective projections that correlate with the classic symptoms of cognitive slowing and decline.
 - Cholinesterase inhibitors enhance synaptic levels of ACh.

Centrally-Acting Cholinesterase Inhibitors

Drug	PK (all oral)	MOA (probable)
Donepezil	• oral; well absorbed; 1x daily dosing • transdermal option (1x/wk) • protein binding >95% • hepatic (minor CYP), active metabolites • 100% bioavailability • Urine mainly / feces: renal dosing not required • t_{1/2} ~70h	reversible • noncompetitive inhibition
Rivastigmine	• 35% bioavailability, cholinesterase hydrolysis, renal (metabolites), duration of activity ~10h	reversible • carbamoylation
Galantamine	• 90% bioavailability; hepatic metabolism (CYP2D6, 3A4); renal, biliary excretion; t_{1/2} 7h	reversible • competitive inhibitor
Tacrine	• no longer used due to hepatotoxicity • oral; t_{1/2} 2-4h; admin 4x daily before meals / inhibits AChE and BuChE; and other effects	

These drugs are generally well tolerated.

Most common side effects: nausea, anorexia, vomiting, diarrhea, altered sleep patterns, vivid dreams

Potentially serious (may require deprescribing):

bradycardia, hypotension, syncope, AV block, unintentional weight loss

- Cholinesterase inhibitors amplify the actions and effects of acetylcholine by inhibiting ACh breakdown, which increases ACh concentration and residence time in cholinergic synapses.
- Cholinesterase inhibitors have both nicotinic and muscarinic effects. Dominant effects vary among organ systems. Cardiovascular, gastrointestinal, eye, and skeletal muscle cholinergic effects are most prominent effects of cholinesterase inhibitors.
- Edrophonium is a simple alcohol that reversibly binds the active site electrostatically and by hydrogen bonds. It has with a very short duration of action.
- Carbamate esters undergo are much more resistant to hydrolysis than ACh. Thus, they have a longer duration of action.
- Organophosphates form an extremely stable phosphate-enzyme bond. Aging further stabilizes bond (irreversible). Organophosphate toxicity can be rapidly fatal.
- Adverse effects of cholinesterase inhibitors reflect the actions of acetylcholine on the various organ systems: diaphoresis, salivation, abdominal pain, nausea, vomiting, diarrhea, urination, bronchoconstriction and secretion of mucus, bradycardia, hypotension, shortness of breath (bronchoconstriction, bronchorrhea), respiratory depression, and respiratory paralysis.
- Atropine is a muscarinic receptor antagonist, which blocks acetylcholine's effects in periphery and CNS.
- Pralidoxime (2-PAM) is a strong nucleophile with a very high affinity for the phosphorus atom in organophosphates that regenerates acetylcholinesterase before aging occurs, but not afterwards. It does not penetrate into the CNS. The two drugs together may reverse both muscarinic and nicotinic signs and symptoms.

Check your knowledge

- A patient who had surgery under general anesthesia facilitated by a nondepolarizing neuromuscular blocker is administered a drug to reverse the neuromuscular blockade. What is the mechanistic rationale for this approach?
- Administration of a carbamate cholinesterase inhibitor (neostigmine or pyridostigmine) reversibly inhibits acetylcholinesterase, which increases ACh synaptic levels. ACh outcompetes the neuromuscular blocker, reversing the blockade.
- A patient diagnosed with myasthenia gravis is given treatment with an oral drug. What is the drug and what is the clinical rationale for this approach?
- Myasthenia gravis is characterized by autoimmune attack on neuromuscular nicotinic receptors, cross-linking them and reducing their numbers. Pyridostigmine inhibits AChE, increasing synaptic ACh levels, which increases the probability of producing an action potential, leading to variable improvement in muscle strength.

Check your knowledge

- The carbamate, organophosphate, Alzheimer's cholinesterase inhibitors have myriad, predictable, mechanism-related adverse effects. What systems are involved?
- All autonomic nervous system functions can be activated because ACh is the neurotransmitter at the nicotinic receptors of the ganglia (PNS and SNS), the nicotinic receptor at the adrenal medulla (SNS), the nicotinic receptors at the neuromuscular junction, and the muscarinic receptors at the effector organs (PNS except sweat glands, SNS).
- Neostigmine and pyridostigmine do not cross the BBB.
- Physostigmine, donepezil, and other Alzheimer disease drugs cross the BBB so have effects on the CNS and peripheral nicotinic and muscarinic receptors.